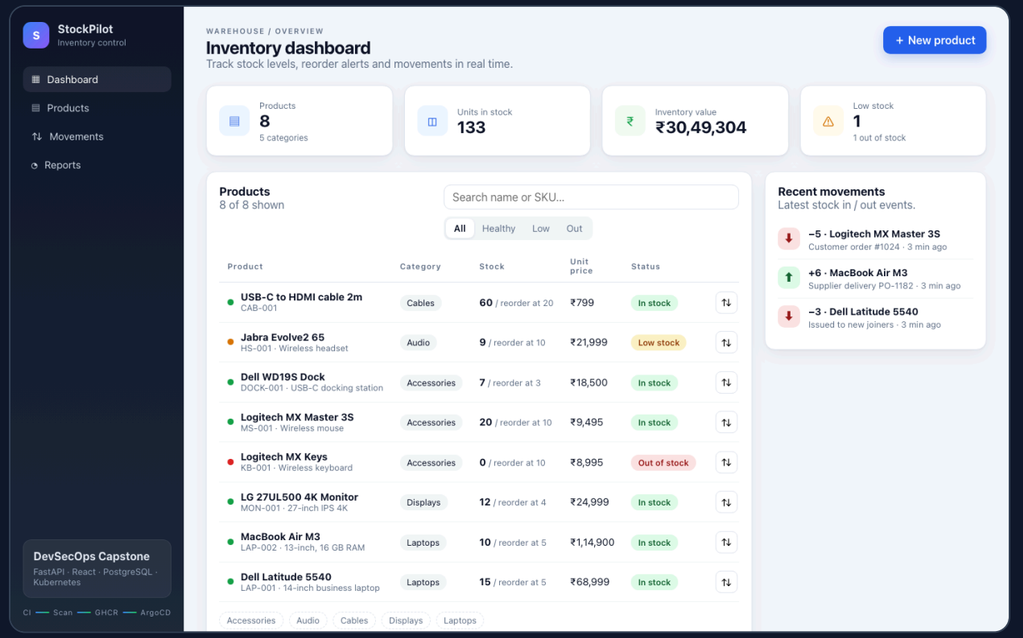
# StockPilot — Inventory Management

## DevSecOps Final Capstone · Gaurav Kumar

github.com/Gaurakumar-s/stockpilot-devsecops

FastAPI · React · PostgreSQL · Docker · GitHub Actions · Trivy · GHCR · Terraform · AWS EKS · Helm · ArgoCD · Prometheus · Grafana




StockPilot dashboard running on AWS EKS

# Project Overview

## PROBLEM

Small warehouses track stock in spreadsheets: no single source of truth and no reorder alerts, so items run out unnoticed.

## SOLUTION — STOCKPILOT

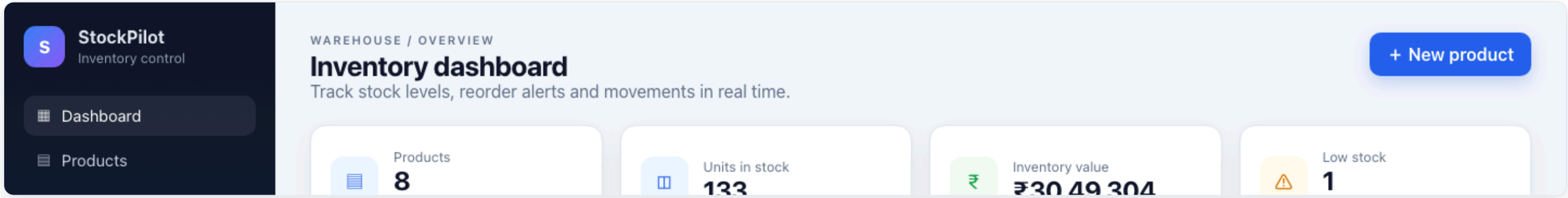
A web app that keeps a live product catalogue, records every stock movement, and flags what needs reordering.

## FEATURES

- Products with SKU, category, quantity, reorder level, unit price
- Stock IN / OUT movements ledger
- Low-stock and out-of-stock alerts
- KPI dashboard: products, units, inventory value, low stock
- Responsive React UI with search and filters

## REST API — 9 ENDPOINTS

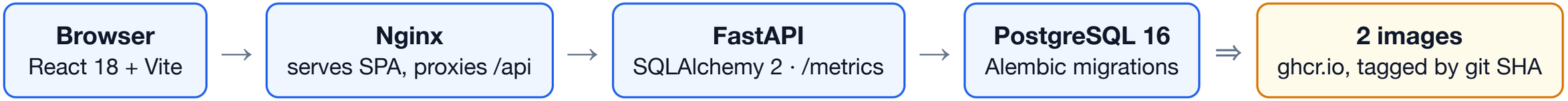
- GET / POST / PUT / DELETE `/api/products`
- GET `/api/products/stats` · GET `/api/products/low-stock`
- POST `/api/products/{id}/movements` · GET `/api/movements`
- Ops: `/health` · `/ready` · `/metrics` · `/docs`



# Tech Stack

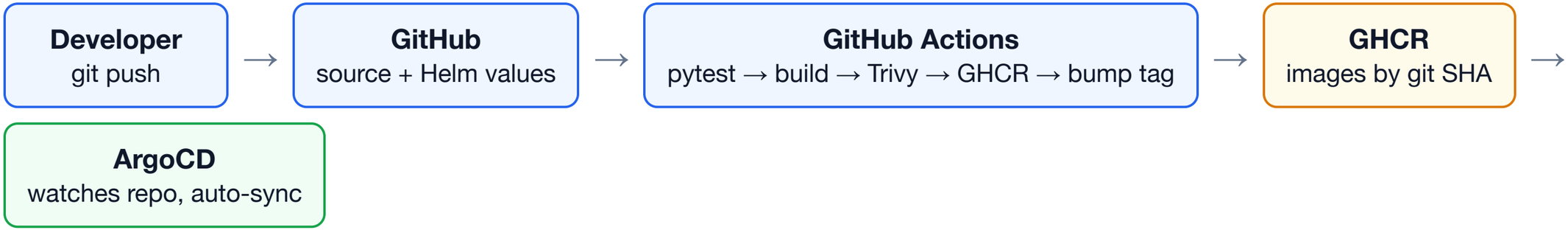
| Application | DevOps / Platform |
|---|---|
| React 18 + Vite (frontend SPA) | Docker — multi-stage builds, non-root containers |
| FastAPI (Python REST API) | Docker Compose (local stack) |
| SQLAlchemy 2 + Alembic (2 migrations) | GitHub Actions (CI/CD) · Trivy (image scanning) · GHCR |
| PostgreSQL 16 | Terraform — AWS VPC + EKS |
| pytest (20 tests) · ruff (lint) | Helm · ArgoCD · ingress-nginx · HPA |
| | Prometheus + Grafana (kube-prometheus-stack) |

## RUNTIME REQUEST PATH

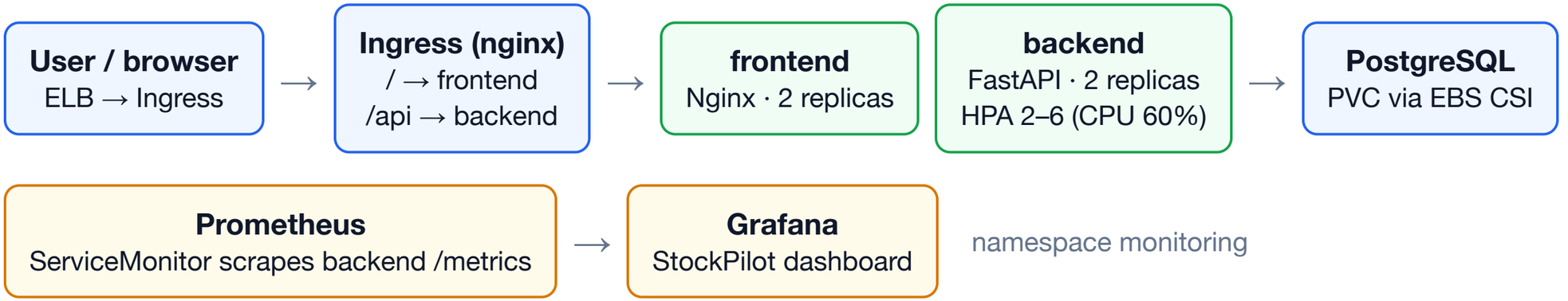


# Architecture

## DELIVERY PATH




## AWS EKS — TERRAFORM-PROVISIONED VPC (2 PUBLIC + 2 PRIVATE SUBNETS, NAT) · NAMESPACE STOCKPILOT



# CI/CD Pipeline — GitHub Actions

Triggers: push to main · pull request · manual (workflow_dispatch)

**1 · backend-test**
ruff lint + pytest (20 tests)
red tests fail the build

→

**2 · frontend-build**
npm ci + vite build

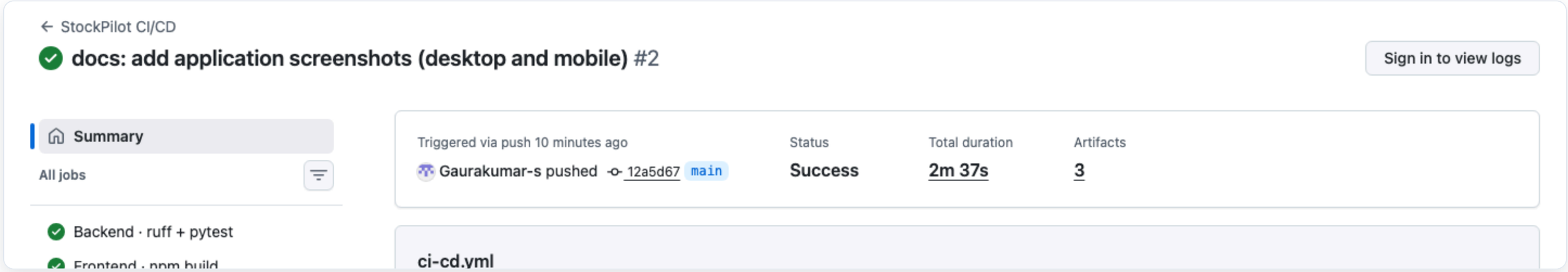
→

**3 · build-scan-push**
docker build both images
Trivy: HIGH/CRITICAL → exit 1
push to GHCR tagged :<git-sha>

→

**4 · update-gitops**
write SHA into helm values
commit → ArgoCD auto-syncs

- Jobs 1 and 2 run in parallel; job 3 needs both green; job 4 needs job 3 and runs only on main.
- Images: `ghcr.io/gaurakumar-s/stockpilot-backend:<git-sha>` and `…-frontend:<git-sha>` — never `latest`, so every pod traces back to one commit.
- Nothing is pushed unless tests and scans pass. The pipeline never runs kubectl: it changes Git, ArgoCD changes the cluster.



# DevSecOps — Security Controls

## TRIVY IMAGE SCANNING IN CI

- Scans OS packages + Python/npm libraries of both images
- Pipeline fails on HIGH / CRITICAL (ignore-unfixed)
- **Finding during build:** starlette 0.46 DoS (CVE-2025-62727) and old Alpine 3.21 packages (musl, zlib, libxml2)
- **Fix:** upgraded to starlette 1.8 / FastAPI 0.143 and nginx-unprivileged on Alpine 3.24

**Scan status: 0 HIGH / 0 CRITICAL**

## OTHER CONTROLS

- Non-root containers: backend uid 10001, frontend uid 101
- securityContext drops ALL capabilities, no privilege escalation
- No secrets in git: .gitignore, terraform.tfvars.example, credentials only in the AWS CLI profile
- Least-privilege GITHUB_TOKEN (packages: write only in the push job)
- Immutable SHA tags; GitOps means no manual cluster access in CI

# Infrastructure — Terraform on AWS

## NETWORK (AP-SOUTHEAST-2)

- VPC 10.20.0.0/16 across 2 Availability Zones
- 2 public subnets — load balancer (ELB)
- 2 private subnets — EKS worker nodes
- Single NAT gateway for node egress

## WORKFLOW

- terraform init → validate → plan → apply → destroy
- terraform.tfvars.example committed; no credentials in git
- ≈ $0.35 / hour — cluster destroyed after the demo

## EKS CLUSTER

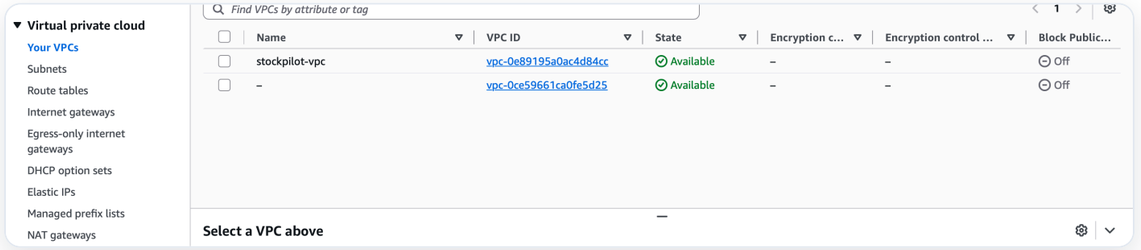
- EKS 1.36, managed node group
- m7i-flex.large × 2 (free-tier eligible, 2 vCPU / 8 GiB; min 2, max 2)
- Add-ons: coredns, kube-proxy, vpc-cni, aws-ebs-csi-driver (EBS policy on the node role) — backs the Postgres PVC

**VPC 10.20.0.0/16**

**public ×2**
ELB + NAT

**private ×2**
EKS nodes

AZ-a and AZ-b each hold one public and one private subnet. Region is fixed by the account's organization policy.




AWS console: stockpilot-vpc in ap-southeast-2

# Kubernetes · Helm · ArgoCD

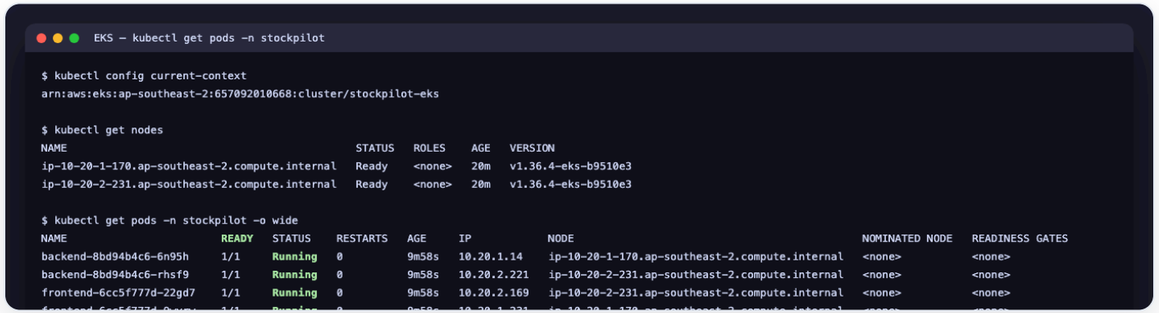
## HELM CHART — HELM/STOCKPILOT

- Deployments: backend ×2, frontend ×2, postgres ×1 + PVC
- ClusterIP Services; Secret for DB credentials
- Ingress (nginx class): / → frontend, /api → backend
- HPA on backend CPU 60 %: 2 → 6 replicas
- ServiceMonitor for Prometheus; liveness + readiness probes
- values-minikube.yaml / values-eks.yaml

## ARGOCD — GITOPS

- Application points at the chart in the repo (main)
- Automated sync + prune + selfHeal
- Git is the source of truth: drift is reverted, deletions pruned

```
$ kubectl get pods,svc,ingress -n stockpilot
$ helm list -n stockpilot
$ kubectl get hpa -n stockpilot
$ argocd app get stockpilot
```






# Observability — Prometheus + Grafana

## METRICS PIPELINE

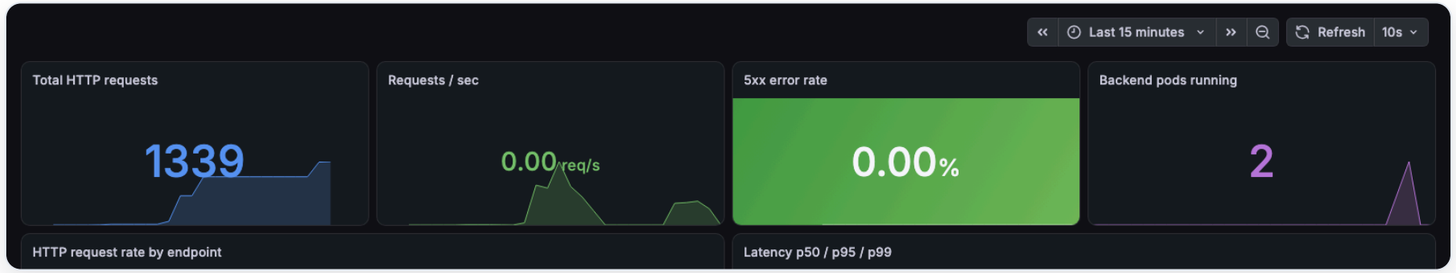
- Backend exposes /metrics via prometheus-fastapi-instrumentator
- http_requests_total, http_request_duration_seconds
- kube-prometheus-stack in namespace monitoring
- ServiceMonitor (release=kube-prometheus-stack) → Targets page: backend UP
- load-test.sh generates traffic for the panels

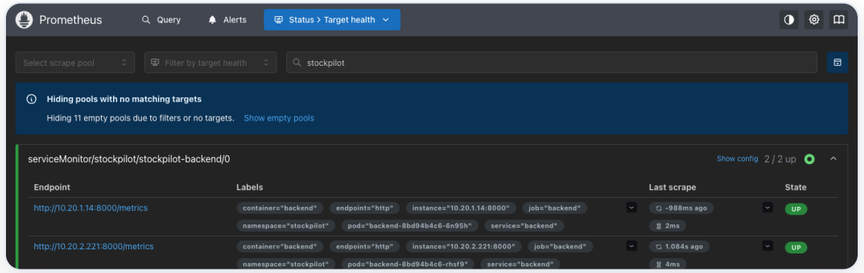
**backend** /metrics → **ServiceMonitor** → **Prometheus** → **Grafana**

## GRAFANA — "STOCKPILOT — APPLICATION METRICS"

| | |
|---|---|
| Requests / sec | p50 / p95 / p99 latency |
| 5xx error rate | Responses by status code |
| Backend CPU (HPA input) | Pods running |

Six panels on one dashboard: traffic, latency, errors, and the CPU signal that drives the HPA.



# Key Learnings & Demo Agenda

## KEY LEARNINGS

- Quality gates before promotion: tests → scan → registry
- Immutable SHA tags give end-to-end traceability
- GitOps removes manual kubectl deploys
- PVCs on EKS need the EBS CSI driver
- Cloud accounts have real constraints: region policy, free-tier instance types, quotas
- Always destroy cloud infrastructure when done

## LIVE DEMO AGENDA

1. Codebase tour
2. Push a commit, watch GitHub Actions
3. GHCR: new SHA-tagged images
4. terraform plan + AWS console
5. ArgoCD: synced application
6. kubectl pods / services / ingress, app via load balancer
7. Prometheus targets
8. Grafana dashboard

**Thank you** · github.com/Gaurakumar-s/stockpilot-devsecops